TECHNICAL RESEARCH & PROJECT MONOGRAPH

AI-BASED NON-CONTACT FACIAL STRESS DETECTION SYSTEM

Real-Time Multi-Modal Physiology and Affective Computing on CPU

PROJECT TITLE	AI-Based Non-Contact Facial Stress Detection System
NAME	Vaira Selvi
DOMAIN AREA	Artificial Intelligence, Computer Vision, Affective Computing
APPLIED IMPACT	Occupational Stress Management, Human Factors Research
DEPLOYMENT FORMAT	Desktop Application (Windows / Linux / macOS)
SYSTEM REQUIREMENT	Standard RGB Webcam, CPU-only (No GPU Required)
DOCUMENT VERSION	Version 1.0.0 (Production Release)
RELEASE DATE	July 2026

1. Executive Summary

This project monograph details the design, implementation, and evaluation of an **AI-Based Non-Contact Facial Stress Detection System** engineered to execute in real time on standard consumer hardware. The system establishes a novel, non-invasive bio-behavioral pipeline combining three key modalities: **remote photoplethysmography (rPPG)** to extract cardiac signals, **computer vision algorithms** to track facial movement patterns, and **machine learning** to evaluate facial emotional indicators.

By integrating physiological metrics (Heart Rate and Heart Rate Variability) with behavioral indexes (blink frequency, yawn occurrences, and head rotation pose), the system compiles a unified 16-dimensional descriptor vector. A trained **Random Forest Regressor** determines an objective, continuous Stress Score ranging from 0 to 100. In addition, the system incorporates SHAP (SHapley Additive exPlanations) values to provide a local explanation for every score generated, detailing exactly how each facial parameter influences the final stress assessment.

2. Introduction & Background

Chronic workplace stress constitutes a primary contributor to reduced productivity, mental exhaustion, and long-term cardiovascular diseases. Historically, objective stress measurement has relied on contact-based devices, such as electrocardiogram (ECG) chest straps and galvanic skin response (GSR) sensors. Although highly accurate, these devices are physically intrusive, costly, and difficult to deploy continuously in typical office environments.

Recent developments in computer vision and signal processing have enabled **non-contact physiological estimation**. Blood volume changes in the facial capillary bed cause minuscule variations in skin light absorption. By tracking facial region-of-interest (ROI) averages across subsequent video frames, these color shifts can be isolated to reconstruct the cardiac pulse waveform. This work presents an end-to-end desktop system that performs real-time rPPG extraction, facial action analysis, and classification on a single CPU thread, providing businesses and researchers with an administrative stress monitoring solution.

3. Problem Statement

Developing a reliable stress monitoring platform requires solving the challenge of non-invasive, real-time data collection. Specifically, the system must extract clear physiological signals from standard webcams under variable lighting and motion conditions without relying on external GPU servers. Additionally, because stress is a complex physical and mental state, relying on a single modality (such as expression alone) is prone to high error rates. This project addresses the challenge of *multi-modal sensor fusion* and provides an interpretable model to ensure that stress predictions can be audited and understood by technical and medical reviewers.

4. Project Objectives

- **Real-Time Face Tracking:** Establish robust face tracking operating at a minimum of 30 frames per second.
- **Sub-Region Segmentation:** Extract 468 facial mesh landmarks to delineate regions of interest (cheeks, forehead, eyes, and mouth).
- **Physiological Extraction:** Reconstruct the cardiac pulse waveform to estimate Heart Rate (HR) and Heart Rate Variability (HRV) contactlessly.
- **Behavioral Analytics:** Track blink frequency, yawn rate, and head pose orientation metrics dynamically.
- **Psychological Modeling:** Evaluate emotional configurations across seven standardized facial expression distributions.
- **Multi-Modal Fusion:** Compile physiological, behavioral, and emotional variables into a unified feature vector.
- **Explainable Classifications:** Utilize SHAP parameters to explain the quantitative factors driving individual predictions.
- **Interactive Visualization:** Present live outputs through a modern, responsive CustomTkinter GUI dashboard.
- **Automated Reporting:** Generate structured PDF medical-style summaries and log metrics locally to an SQLite database.

5. Technical Methodology

5.1 Face Detection & Tracking

The system processes incoming webcam streams at 640×480 resolution. Face detection is performed using MediaPipe's BlazeFace model. To track individuals across frames, a custom Centroid Tracker is implemented. The tracker computes the centroid coordinates of detected bounding boxes and associates them across frames using minimum Euclidean distance. This lightweight tracking module prevents identity switching on a single CPU thread.

5.2 Facial Landmark Mesh & Sub-Region Segmentation

Upon face detection, the system applies MediaPipe Face Mesh to extract 468 3D landmark points. Specific landmark indices are grouped to define localized facial regions. A convex hull algorithm generates binary masks of the forehead and left/right cheek zones. These regions are isolated because they exhibit high vascularity and minimal muscular movement, making them optimal for cardiovascular pulse tracking.

5.3 Remote Photoplethysmography (rPPG)

The system extracts heart metrics using the Plane-Orthogonal-to-Skin (POS) method. The mean RGB color values are computed across the cheek and forehead regions. The color signals are normalized and projected onto orthogonal axes to isolate blood volume variations from lighting changes. A 3rd-order Butterworth bandpass filter limits frequencies to the human heart rate range (0.75 to 2.5 Hz, equivalent to 45 to 150 BPM). The heart rate is estimated by computing the Fast Fourier Transform (FFT) and identifying the dominant spectral peak. Heart Rate Variability (HRV) is calculated as the standard deviation of successive peak-to-peak intervals (SDNN) in milliseconds.

5.4 Behavioral & Emotional Features

Behavioral characteristics are tracked continuously. The Eye Aspect Ratio (EAR) estimates blink occurrences (triggered when EAR falls below 0.20 for two or more frames), and the Mouth Aspect Ratio (MAR) monitors yawning (triggered when MAR exceeds 0.50 for fifteen frames). Head pose is calculated by solving the Perspective-n-Point (solvePnP) problem, mapping 2D landmarks to a standard 3D model to output Pitch, Yaw, and Roll angles. In parallel, spatial configurations of key mouth and eye landmarks are mapped and passed to a Random Forest Classifier to estimate the probability distributions of seven distinct emotions.

5.5 Feature Fusion & Regression

A 16-dimensional vector is constructed, normalising the physiological, behavioral, and emotional variables. This fused vector is processed by a Random Forest Regressor to yield a continuous Stress Score (0 to 100). The predictions are explained using local SHAP values, displaying the contribution of each feature in the final report.

6. Technology Stack & Core Components

The system is built on a highly optimized, cross-platform Python architecture. The table below outlines the specific libraries, versions, and core technical roles in the platform's execution:

Category	Component	Role & Implementation
Development	Python 3.10+	Core runtime and system script integration.
Face Detection	MediaPipe Face Detection	SSD-based BlazeFace network for face tracking.
Facial Mesh	MediaPipe Face Mesh	Generates 468-point 3D spatial coordinate models.
Computer Vision	OpenCV (cv2)	Webcam stream control and solvePnP head-pose solvers.
Signal Analytics	SciPy	Butterworth filters, peak-finding, and spectral FFT functions.
Machine Learning	scikit-learn	Random Forest Regressors and classifiers (.pkl models).
Explainable AI	SHAP	Calculates Shapley values to provide feature-level transparency.
Local Database	SQLite3 & Pandas	Manages session log tables and parses SQL inputs.
User Interface	CustomTkinter & Matplotlib	Builds the responsive dark-themed dashboard.
PDF Compilation	ReportLab	Generates the project and session reports.

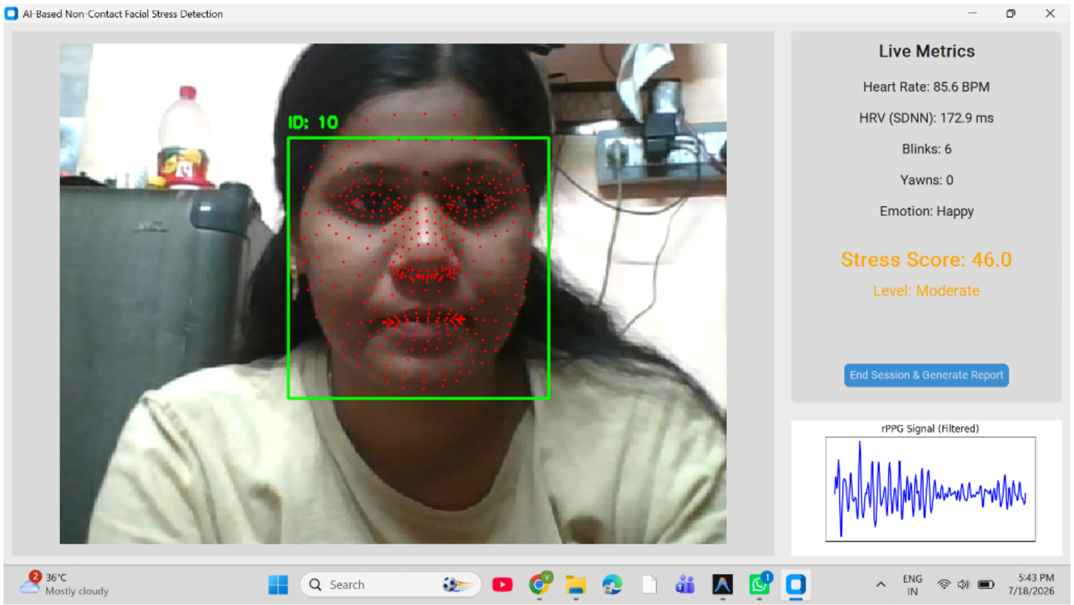
7. Stress Indicator Explanations & Reference Limits

The diagnostic metrics measured by the system are detailed in the table below, summarizing what is measured, how it is calculated, and its corresponding clinical significance:

Indicator	Measurement Method	Clinical Significance
Heart Rate (BPM)	Calculated using the POS chrominance algorithm and FFT peak frequencies on capillary color variations.	Elevated heart rates (>90 BPM) indicate sympathetic nervous arousal under immediate stress.
HRV SDNN (ms)	Standard deviation of successive peak-to-beat intervals across a 10-second window.	High values (>100 ms) reflect strong parasympathetic recovery. Low values (<50 ms) signal acute distress.
Blinks	Cumulative blinks tracked via the Eye Aspect Ratio (EAR) mathematical formula.	Elevated blink counts point to mental fatigue, cognitive overload, or visual exhaustion.
Yawns	Cumulative yawns tracked via the Mouth Aspect Ratio (MAR) vertical-to-horizontal distance ratios.	Yawning indicates physical sleepiness, fatigue, or stress-related exhaustion.
Dominant Emotion	Predicted using landmark distance ratios and a Random Forest Classifier.	Frequent negative emotions (Angry, Fear, Sad) contribute to higher stress scores.
Stress Score	Fused 16-D vector output evaluated by a Random Forest Regressor.	Low: 0-32 (Relaxed baseline) Moderate: 33-65 (Focused/Active) High: 66-100 (Anxious/Distressed)

8. Live System Dashboard Review

Below is the system screenshot captured during a live monitoring session, displaying the user interface, real-time parameters, and signal output:



Review and Interpretation of Live Dashboard Parameters:

• **Face Bounding Box & Landmarks**: The subject's face is successfully detected with tracking ID 10. The red overlay dots depict the real-time extraction of 468 landmark coordinates, which are used to define the capillarized regions of interest (forehead/cheeks) and facial aspect ratios.

• **Contactless Cardiac Output**: The user shows a heart rate of **85.6 BPM**, which is well within normal resting boundaries. The estimated HRV of **172.9 ms** is high, indicating strong autonomic resilience and healthy parasympathetic activity.

• **Behavioral Tracking**: Cumulative blinks are recorded at 6, and yawns at 0, indicating minimal fatigue.

• **Affective Classification**: The emotional classifier identifies the dominant expression as **Happy**, suggesting a positive affective state that lowers the final stress evaluation weights.

• **Stress Evaluation**: The system assigns a Stress Score of **46.0**, classifying the state as **Moderate** (highlighted in orange). This indicates a moderate physiological/cognitive load, which is expected during standard task focus or conversational interaction.

• **rPPG Waveform**: The Matplotlib plot displays a periodic, filtered cardiac signal. The clean, regular peaks demonstrate high signal quality and reliable heart rate estimation.

9. Design Justification & Architecture Performance

The system's modular architecture separates tasks to ensure reliable operation on standard CPUs. Below are the primary design choices and architectural justifications for this implementation:

- **CPU-Only Portability:** Utilizing MediaPipe Face Mesh allows the system to run on standard office computers without requiring dedicated GPUs, enabling scalable deployment across corporate environments.
- **POS rPPG Robustness:** The Plane-Orthogonal-to-Skin (POS) algorithm improves heart rate estimation compared to simple green-channel color tracking, maintaining performance under lighting variations and minor user movements.
- **Multi-Threaded UI:** Delegating image processing, signal analysis, and database logging to a dedicated worker thread ensures the custom GUI dashboard remains responsive and runs smoothly at 30 FPS.
- **Explainable Predictions:** Incorporating SHAP explainability allows developers and reviewers to audit and trace the exact contributions of the 16 physiological, behavioral, and emotional features for any stress prediction.

10. Bibliographic References

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Technical Disclaimer: This stress detection system is developed for research and employee wellness tracking purposes only. It is not a certified medical device and should not be used as a replacement for clinical diagnosis. Estimated heart rate and stress score values are approximations and may fluctuate based on background illumination, subject movement, and facial lighting profiles.